

« Experiments

AI Experiments



From left to right: George R. Perreault, head of the Library of Congress Data Processing Office, standing at the computer storage unit; Ernest Acosta Jr., digital computer programmer, working at the card reader unit; and Joseph B. Murphy, digital computer programmer, inserting a new tape in one of the tape units. Jan. 20, 1964. Item 1333, box 69, Photographs, Illustrations & Objects, Library of Congress Archives, Manuscript Division, Library of Congress, Washington, D.C. Read more about the history of computing at the Library in this [blog post](#).

Experimenting with artificial intelligence and machine learning at the Library of Congress.

The Library of Congress researches and experiments with artificial intelligence (AI) and machine learning (ML), focusing on ethical uses of these technologies, and addressing the challenges of their adoption in libraries and cultural memory organizations.

Below are examples of the research and experiments we have undertaken into how these technologies can enhance collections, operations, and services. This page also includes information about the LC Labs Artificial Intelligence Planning Framework, developed using the lessons we learned during our early AI and machine learning experimentation.

For more information about how AI is being used across the Library of Congress, please visit <https://loc.gov/digital-strategy/artificial-intelligence>.

Experiments

- [Speech to Text Viewer](#): proof of concept tool testing off-the-shelf transcription tools
- [Exploring ML with the Project Aida team](#): six explorations of how machine learning could be applied to the Library's digital collections
- [Experimental Access](#): exploring experimental ways of providing access to the Library's digital collections
- [Humans-in-the-Loop](#): an experimental humans in the loop workflow for pairing human decision-making with automated processes
- [Exploring Computational Description](#): investigating how machine learning can help with cataloging

- [Newspaper Navigator](#) by 2020 Innovator in Residence Ben Lee
- [Citizen DJ](#) by 2020 Innovator in Residence Brian Foo
- [America's Public Bible: Machine-Learning Detection of Biblical Quotations Across LOC Collections via Cloud Computing](#)
by research expert Lincoln Mullen
- [Access & Discovery of Documentary Images](#) by research expert Lauren Tilton
- [Situating Ourselves in Cultural Heritage: Using Neural Nets to Expand the Reach of Metadata and See Cultural Data on Our Own Terms](#) by research expert Andromeda Yelton

Reports

- [The Machine Learning + Libraries Summit: Event Summary \(2020\)](#) includes detailed information about the Machine Learning + Libraries Summit hosted by LC Labs in September 2019
- [Digital Libraries, Intelligent Data Analytics, and Augmented Description: Final Report \(2020\)](#) details exploratory projects conducted by the Project Aida Team at the University of Nebraska Lincoln and addresses social and technical challenges for the development of ML in the cultural heritage sector
- [Machine Learning + Libraries: A Report on the State of the Field \(2020\)](#) Ryan Cordell, Associate Professor of English at Northeastern University reports on the "state of the field in machine learning and libraries"
- [Feasible, Adaptable and Shared: A Call for a Community Framework for Implementing ML and AI" \(2022\)](#) by Abigail Potter, Meghan Ferriter, Eileen J. Manchester, and Jaime Mears. *Proceedings of the 18th International Conference on Digital Preservation 2022*, p. 145

LC Labs Artificial Intelligence Planning Framework

LC Labs has published a [planning framework](#) to support the responsible exploration and potential adoption of AI in cultural heritage organizations. The framework includes three planning phases Understand, Experiment and Implement, each of which enables identification and evaluation of the data, models and people necessary for applying AI technologies responsibly and effectively. We've developed a set of worksheets, questionnaires, and workshops to engage stakeholders and staff and identify priorities for future AI

experiments, and we continue to refine and update these resources as they guide a variety of uses of AI.

Access the Planning Framework on [GitHub](#) and read more about it in this [post on the Signal Blog](#).

Worldwide Community Engagement

LC Labs regularly engages in international conversations about AI, representing a deep engagement with questions about the use of AI in libraries, archives and museums. The Library is a founding member organization of the international AI for Libraries, Archives and Museums (AI4LAM) organization, and we collaborate with other national libraries and research institutions in the International Federation of Library Associations (IFLA). Staff have also joined federal communities of practice including the Equitable Data Community of Practice, and the General Services Administration (GSA) AI Community of Practice, and founded the DC Chapter of AI4LAM.

These community collaborations continue to inspire and deepen AI experimentation at the Library and support new AI research in the wider field of libraries, archives and museums. There are a number of areas where we see potential for AI to make a positive difference in our core work. Our aims for research and experimentation include:

- Deriving machine-readable text from digitized documents using Optical Character Recognition (OCR) and related technologies
- Generating data for bibliographic records and creating standardized catalog records
- Developing content provenance and authenticity standards and practices
- Producing machine-readable data from historic records
- Extracting geographic place and subject terms from collection data
- Supporting metadata generation workflows across formats, including audio visual and born-digital archives
- Refining the use of natural language processing (NLP) and entity management
- Developing effective AI literacy and learning resources tailored for libraries, museums, and archives

We are committed to measuring the quality of outcomes, testing humans-in-the-loop approaches and analyzing the use of AI in larger workflows. Our processes for experimentation and research in AI reflect our interest in the above areas to support key components of the Library's mission, including supporting digital research, analysis and and discovery services that uphold high standards for authenticity, accuracy, effectiveness, and efficiency.



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Questions? Contact LC Labs at LC-Labs@loc.gov